

Software Specification:

Operating System: Windows 10/11, Linux (Ubuntu 18.04 or later), or macOS (Latest versions preferred)

Programming Language: Python 3.8 or above

Libraries and Frameworks:

OpenCV (for video and image processing)

PyTorch or TensorFlow (for deep learning and running YOLOv8)

YOLOv8 (Ultralytics repository)

Text-to-Speech library (like pyttsx3 or gTTS)

NumPy (for mathematical operations)

Additional Python packages: matplotlib.

Other Tools:

IDE: VS Code, PyCharm, or Jupyter Notebook for development

Virtual Environment: (Optional) Anaconda or venv for managing dependencies easily